occultation, this finding (and much subsequent work) has led to the discovery of nine very thin, very dark circum-planetary rings, giving Uranus the appearance of a bull's-eye in the sky.

Surrounding the rings, Earthbound observers understood, were the concentric orbits of the five moons then known: Miranda, Ariel, Umbriel, Titania, and Oberon. They're named after characters in Shakespeare's *A Midsummer Night's Dream* and *The Tempest*, and in Alexander Pope's *The Rape of the Lock*. Two of them were found by Herschel himself. The innermost of the five, Miranda, was discovered as recently as 1948, by my teacher G. P. Kuiper.* I remember how great an achievement the discovery of a new moon of Uranus was considered back then. The near-infrared light reflected by all five moons subsequently revealed the spectral signature of ordinary water ice on their surfaces. And no wonder—Uranus is so far from the Sun that it is no brighter there at noontime than it is after sunset on Earth. The temperatures are frigid. Any water must be frozen.

A REVOLUTION IN OUR UNDERSTANDING of the Uranus system—the planet, its rings, and its moons—began on January 24, 1986. On that day, after a journey of 8½ years, the *Voyager 2* spacecraft sailed very near to Miranda, and hit the bull's-eye in the sky. Uranus' gravity then flung it on to Neptune. The spacecraft returned 4,300 close-up pictures of the Uranus system and a wealth of other data.

Uranus was found to be surrounded by an intense radiation belt, electrons and protons trapped by the planet's magnetic field. *Voyager* flew through this radiation belt, measuring the magnetic field and the trapped charged particles as it went. It also detected—in changing timbres, harmonies, and nuance, but mainly in *fortissimo*—a cacophony of radio waves generated by the speeding, trapped particles. Something similar was discovered on Jupiter and Saturn and would be later found at Neptune—but always with a theme and counterpoint characteristic of each world.

On Earth the magnetic and geographical poles are quite close together. On Uranus the magnetic axis and the axis of rotation are